第四章　積分

4.1 不定積分

定義 (反導函數)**.** 給定 $F(x)$，若 $\dfrac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x)$，則稱 $F(x)$ 為 $f(x)$ 的反導函數（antiderivative）。

性質. 若 $F(x)$，$G(x)$ 分別為 $f(x)$，$g(x)$ 的反導函數，$c \in \mathbb{R}$。則

- $F(x) + c$ 為 $f(x)$ 的反導函數。
- $c\,F(x)$ 為 $c\,f(x)$ 的反導函數。
- $F(x) + G(x)$ 為 $f(x) + g(x)$ 的反導函數。

結論.

- $\dfrac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x) \implies \mathrm{d}F(x) = f(x)\cdot\mathrm{d}x \implies F(x) = \displaystyle\int f(x)\cdot\mathrm{d}x = \int f(x)\,\mathrm{d}x$
- $F(x)$ 為 $f(x)$ 的反導函數 $\iff$ $f(x)$ 的反導函數為 $F(x)$ $\iff$ $F(x)$ 的導函數為 $f(x)$ $\iff$ $F(x)$（對 x）的微分為 $f(x)$ $\iff$ $f(x)$（對 x）的（不定）積分為 $F(x)$
- $f(x)$ 的反導函數 $\equiv f(x)$（對 x）的（不定）積分
- 基礎積分集：以下 $\alpha \neq -1$，$a \in \mathbb{R}\setminus\{0\}$。

$f(x)$	x^α	$\dfrac{1}{x}$	e^{ax}	$\sin ax$	$\cos ax$	$\dfrac{1}{\sqrt{a^2-x^2}}$	$\dfrac{1}{a^2+x^2}$
$\displaystyle\int f(x)\,\mathrm{d}x$	$\dfrac{1}{\alpha+1}x^{\alpha+1}$	$\ln\lvert x\rvert$	$\dfrac{1}{a}e^{ax}$	$-\dfrac{1}{a}\cos ax$	$\dfrac{1}{a}\sin ax$	$\sin^{-1}\dfrac{x}{\lvert a\rvert}$	$\dfrac{1}{a}\tan^{-1}\dfrac{x}{a}$

- (Liouville) e^{-x^2}，$\dfrac{\sin x}{x}$，x^x 無（初等函數形式之）反導函數！

例. $\displaystyle\int \sqrt{x}\,\mathrm{d}x = \frac{2}{3}x^{\frac{3}{2}} + c$，$\displaystyle\int x^\pi\,\mathrm{d}x = \frac{1}{\pi+1}x^{\pi+1} + c$，$\displaystyle\int \sin\pi x\,\mathrm{d}x = -\frac{1}{\pi}\cos\pi x + c$，$\displaystyle\int \cos xy\,\mathrm{d}x = \frac{1}{y}\sin xy + c$，$\displaystyle\int e^{-2x}\,\mathrm{d}x = -\frac{1}{2}e^{-2x} + c$，$\displaystyle\int \frac{1}{\sqrt{\pi - x^2}}\,\mathrm{d}x = \sin^{-1}\frac{x}{\sqrt{\pi}} + c$，$\displaystyle\int \frac{1}{e+u^2}\,\mathrm{d}u = \frac{1}{\sqrt{e}}\tan^{-1}\frac{u}{\sqrt{e}} + c$

變數變換法

結論. $\dfrac{\mathrm{d}}{\mathrm{d}x}f(g(x)) = f'(g(x))\cdot g'(x) \implies \mathrm{d}f(g(x)) = f'(g(x))\cdot g'(x)\,\mathrm{d}x \implies f(g(x)) = \displaystyle\int f'(g(x))\cdot g'(x)\,\mathrm{d}x$。令 $u = g(x)$，則 $\dfrac{\mathrm{d}}{\mathrm{d}x}u = g'(x) \implies \mathrm{d}u = g'(x)\,\mathrm{d}x$；故 $\displaystyle\int f'(g(x))\cdot g'(x)\,\mathrm{d}x = \int f'(u)\,\mathrm{d}u = f(u) + c = f(g(x)) + c$。

例. 求 $\displaystyle\int \frac{x}{x^2+1}\,\mathrm{d}x$。

解. 令 $u = x^2 + 1$，則 $\mathrm{d}u = 2\,x\,\mathrm{d}x \implies x\,\mathrm{d}x = \dfrac{1}{2}\mathrm{d}u$。故 $\displaystyle\int \frac{x}{x^2+1}\,\mathrm{d}x = \int \frac{1}{x^2+1}\cdot x\,\mathrm{d}x = \int \frac{1}{u}\cdot\frac{1}{2}\,\mathrm{d}u = \frac{1}{2}\ln\lvert u\rvert + c = \frac{1}{2}\ln(x^2+1) + c$。

例. 求 $\displaystyle\int \frac{\sin(3\ln x)}{x}\,\mathrm{d}x$。

解. 令 $u=\ln x$，則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$。故 $\displaystyle\int\frac{\sin(3\ln x)}{x}\,\mathrm{d}x=\int\sin(3\ln x)\cdot\frac{1}{x}\,\mathrm{d}x=\int\sin 3u\cdot\mathrm{d}u=-\frac{1}{3}\cos 3u+c=-\frac{1}{3}\cos(3\ln x)+c$。

例. 求 $\displaystyle\int e^x\sqrt{1+e^x}\,\mathrm{d}x$。

解. 令 $u=1+e^x$, 則 $\mathrm{d}u=e^x\,\mathrm{d}x$。故 $\displaystyle\int e^x\sqrt{1+e^x}\,\mathrm{d}x=\int\sqrt{1+e^x}\cdot e^x\,\mathrm{d}x=\int\sqrt{u}\cdot\mathrm{d}u=\frac{u^{\frac{3}{2}}}{\frac{3}{2}}+c=\frac{2}{3}(1+e^x)^{\frac{3}{2}}+c$

例. 求 $\displaystyle\int\frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x$。

解. 令 $u=e^x$，則 $\mathrm{d}u=e^x\,\mathrm{d}x$。故 $\displaystyle\int\frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x=\int\frac{1}{\sqrt{2-u^2}}\,\mathrm{d}u=\sin^{-1}\frac{u}{\sqrt{2}}+c=\sin^{-1}\frac{e^x}{\sqrt{2}}+c$。

例. 求 $\displaystyle\int\frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x$。

解. $\displaystyle\int\frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x=\int\frac{1}{e^x\sqrt{1-e^{-2x}}}\,\mathrm{d}x=\int\frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x$。令 $u=e^{-x}$，則 $\mathrm{d}u=-e^{-x}\,\mathrm{d}x\implies e^{-x}\,\mathrm{d}x=-\mathrm{d}u$；故 $\displaystyle\int\frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x=\int\frac{1}{\sqrt{1-e^{-2x}}}\cdot e^{-x}\,\mathrm{d}x=-\int\frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u=-\sin^{-1}u+c=-\sin^{-1}e^{-x}+c$。

例. 求 $\displaystyle\int\frac{1}{x^2+4x+5}\,\mathrm{d}x$。

解. $\displaystyle\int\frac{1}{x^2+4x+5}\,\mathrm{d}x=\int\frac{1}{(x+2)^2+1}\,\mathrm{d}x$。令 $u=x+2$, 則 $\mathrm{d}u=\mathrm{d}x$；故 $\displaystyle\int\frac{1}{(x+2)^2+1}\,\mathrm{d}x=\int\frac{1}{u^2+1}\,\mathrm{d}u=\tan^{-1}u+c=\tan^{-1}(x+2)+c$。

例. 求 $\displaystyle\int\frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x$

解. $\displaystyle\int\frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x=\int\frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x$。令 $u=x-1$，則 $\mathrm{d}u=\mathrm{d}x$；故 $\displaystyle\int\frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x=\int\frac{1}{\sqrt{5-u^2}}\,\mathrm{d}u=\sin^{-1}\frac{u}{\sqrt{5}}+c=\sin^{-1}\frac{x-1}{\sqrt{5}}+c$。

例. 求 $\displaystyle\int\tan x\,\mathrm{d}x$。

解. $\displaystyle\int\tan x\,\mathrm{d}x=\int\frac{\sin x}{\cos x}\,\mathrm{d}x$。令 $u=\cos x$，則 $\mathrm{d}u=-\sin x\,\mathrm{d}x$；故 $\displaystyle\int\frac{\sin x}{\cos x}\,\mathrm{d}x=\int\frac{-1}{u}\,\mathrm{d}u=-\ln|u|+c=-\ln|\cos x|+c=\ln|\sec x|+c$

例. 求 $\displaystyle\int\sec x\,\mathrm{d}x$。

解. $\displaystyle\int\sec x\,\mathrm{d}x=\int\frac{\sec x\cdot(\sec x+\tan x)}{\sec x+\tan x}\,\mathrm{d}x$。令 $u=\sec x+\tan x$，則 $\mathrm{d}u=(\sec^2 x+\sec x\tan x)\,\mathrm{d}x$；故 $\displaystyle\int\frac{\sec x\cdot(\sec x+\tan x)}{\sec x+\tan x}\,\mathrm{d}x=\int\frac{1}{u}\,\mathrm{d}u=\ln|u|+c=\ln|\sec x+\tan x|+c$

例. 求 $\displaystyle\int\cos^5 ax\,\mathrm{d}x$。

解. $\displaystyle\int\cos^5 ax\,\mathrm{d}x=\int(1-\sin^2 ax)^2\cos ax\,\mathrm{d}x$。令 $u=\sin ax$, 則 $\mathrm{d}u=a\cos ax\,\mathrm{d}x$；故 $\displaystyle\int(1-\sin^2 ax)^2\cos ax\,\mathrm{d}x=\frac{1}{a}\int(1-u^2)^2\,\mathrm{d}u=\frac{1}{a}\int(1-2u^2+u^4)\,\mathrm{d}u=\frac{1}{a}\left(u-\frac{2}{3}u^3+\frac{1}{5}u^5\right)+c=\frac{1}{a}\left(\sin ax-\frac{2}{3}\sin^3 ax+\frac{1}{5}\sin^5 ax\right)+c$。

例. 求 $\int \cos^4 ax\,\mathrm{d}x$。

解. $\int \cos^4 ax\,\mathrm{d}x = \int \left(\frac{1+\cos 2ax}{2}\right)^2 \mathrm{d}x = \frac{1}{4}\int \left(1+2\cos 2ax+\cos^2 2ax\right)\mathrm{d}x = \frac{1}{4}\int \left(1+2\cos 2ax+\frac{1+\cos 4ax}{2}\right)\mathrm{d}x = \int \left(\frac{3}{8}+\frac{1}{2}\cos 2ax+\frac{1}{8}\cos 4ax\right)\mathrm{d}x = \frac{3}{8}x+\frac{1}{4a}\sin 2ax+\frac{1}{32a}\sin 4ax+c$

習題. 以變數變換法求下列不定積分。注意：可能會因為常數項而跟此處答案不同。

1. $\int e^{2x}\sin e^{2x}\,\mathrm{d}x = -\frac{1}{2}\cos e^{2x}+c$
2. $\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x = -e^{-\frac{x^2}{2}}+c$
3. $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x = -2\cos\sqrt{x}+c$
4. $\int x^2 2^{x^3+1}\,\mathrm{d}x = \frac{2^{x^3+1}}{3\ln 2}+c$
5. $\int \frac{\ln x}{x}\,\mathrm{d}x = \frac{1}{2}(\ln x)^2+c$
6. $\int \frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x = \sqrt{x^2+2x+3}+c$
7. $\int \frac{x^2}{2+x^6}\,\mathrm{d}x = \frac{1}{3\sqrt{2}}\tan^{-1}\frac{x^3}{\sqrt{2}}+c$
8. $\int \frac{1}{e^x+e^{-x}}\,\mathrm{d}x = \tan^{-1}e^x+c$
9. $\int \frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x = -\sqrt{1-x^2}+\sin^{-1}x+c$
10. $\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,\mathrm{d}x = \sin^{-1}(\tan x)+c$
11. $\int \sin^4 x\cos^5 x\,\mathrm{d}x = \frac{\sin^5 x}{5}-\frac{2\sin^7 x}{7}+\frac{\sin^9 x}{9}+c$
12. $\int \sin^2 x\cos^2 x\,\mathrm{d}x = \frac{x}{8}-\frac{\sin 4x}{32}+c$
13. $\int \sin^{-\frac{2}{3}} x\cos^3 x\,\mathrm{d}x = 3\sin^{\frac{1}{3}} x-\frac{3}{7}\sin^{\frac{7}{3}} x+c$
14. $\int \frac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x = \frac{\sin^4(\ln x)}{4}-\frac{\sin^6(\ln x)}{6}$
15. $\int \frac{\sin^3 x}{\cos^4 x}\,\mathrm{d}x = -\frac{1}{\cos x}+\frac{1}{3\cos^3 x}+c$
16. $\int \cos ax\cos bx\,\mathrm{d}x = \frac{\sin(a-b)x}{2(a-b)}+\frac{\sin(a+b)x}{2(a+b)}+c$
17. $\int \sin ax\sin bx\,\mathrm{d}x = \frac{\sin(a-b)x}{2(a-b)}-\frac{\sin(a+b)x}{2(a+b)}+c$
18. $\int \sin ax\cos bx\,\mathrm{d}x = -\frac{\cos(a-b)x}{2(a-b)}-\frac{\cos(a+b)x}{2(a+b)}$

解.

1. 令 $u=e^{2x}$, 則 $\mathrm{d}u = 2e^{2x}\,\mathrm{d}x \implies e^{2x}\,\mathrm{d}x = \frac{1}{2}\mathrm{d}u$。故 $\int e^{2x}\sin e^{2x}\,\mathrm{d}x = \int \sin e^{2x}\cdot e^{2x}\,\mathrm{d}x = \int \sin u\cdot\frac{1}{2}\mathrm{d}u = -\frac{1}{2}\cos u+c = -\frac{1}{2}\cos e^{2x}+c$。
2. 令 $u=\frac{x^2}{2}$, 則 $\mathrm{d}u = x\,\mathrm{d}x$, 故 $\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x = \int e^{-\frac{x^2}{2}}\cdot x\,\mathrm{d}x = \int e^{-u}\,\mathrm{d}u = -e^{-u}+c = -e^{-\frac{x^2}{2}}+c$
3. 令 $u=\sqrt{x}$, 則 $\mathrm{d}u = \frac{1}{2\sqrt{x}}\mathrm{d}x \implies \frac{1}{\sqrt{x}}\mathrm{d}x = 2\,\mathrm{d}u$, 故 $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x = \int \sin\sqrt{x}\cdot\frac{1}{\sqrt{x}}\,\mathrm{d}x = \int \sin u\cdot 2\,\mathrm{d}u = -2\cos u+c = -2\cos\sqrt{x}+c$
4. 令 $u=x^3+1$, 則 $\mathrm{d}u = 3x^2\,\mathrm{d}x \implies x^2\,\mathrm{d}x = \frac{1}{3}\mathrm{d}u$, 故 $\int x^2 2^{x^3+1}\,\mathrm{d}x = \int 2^{x^3+1}\cdot x^2\,\mathrm{d}x = \int 2^u\cdot\frac{1}{3}\mathrm{d}u = \frac{1}{3}\int e^{u\ln 2}\,\mathrm{d}u = \frac{1}{3\ln 2}e^{u\ln 2}+c = \frac{2^{x^3+1}}{3\ln 2}+c$
5. 令 $u=\ln x$, 則 $\mathrm{d}u = \frac{1}{x}\mathrm{d}x$, 故 $\int \frac{\ln x}{x}\,\mathrm{d}x = \int \ln x\cdot\frac{1}{x}\,\mathrm{d}x = \int u\,\mathrm{d}u = \frac{1}{2}u^2+c = \frac{1}{2}(\ln x)^2+c$
6. 令 $u=x^2+2x+3$, 則 $\mathrm{d}u = (2x+2)\,\mathrm{d}x \implies (x+1)\,\mathrm{d}x = \frac{1}{2}\mathrm{d}u$, 故 $\int \frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x = \int \frac{1}{\sqrt{x^2+2x+3}}\cdot(x+1)\,\mathrm{d}x = \int \frac{1}{\sqrt{u}}\cdot\frac{1}{2}\mathrm{d}u = \sqrt{u}+c = \sqrt{x^2+2x+3}+c$

7. 令 $u=x^3$，則 $\mathrm{d}u=3x^2\,\mathrm{d}x \implies x^2\,\mathrm{d}x=\dfrac{1}{3}\,\mathrm{d}u$，故 $\displaystyle\int\frac{x^2}{2+x^6}\,\mathrm{d}x=\int\frac{1}{2+x^6}\cdot x^2\,\mathrm{d}x=\int\frac{1}{2+u^2}\cdot\frac{1}{3}\,\mathrm{d}u=\frac{1}{3\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}}+c=\frac{1}{3\sqrt{2}}\tan^{-1}\frac{x^3}{\sqrt{2}}+c$

8. $\displaystyle\int\frac{1}{e^x+e^{-x}}\,\mathrm{d}x=\int\frac{e^x}{e^{2x}+1}\,\mathrm{d}x$。令 $u=e^x$，則 $\mathrm{d}u=e^x\,\mathrm{d}x$，故 $\displaystyle\int\frac{e^x}{e^{2x}+1}\,\mathrm{d}x=\int\frac{1}{e^{2x}+1}\cdot e^x\,\mathrm{d}x=\int\frac{1}{u^2+1}\,\mathrm{d}u=\tan^{-1}u+c=\tan^{-1}e^x+c$。

9. $\displaystyle\int\frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x=\int\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x+\int\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$。令 $u=\sqrt{1-x^2}$,則 $\mathrm{d}u=\dfrac{-x}{\sqrt{1-x^2}}\,\mathrm{d}x \implies \dfrac{x}{\sqrt{1-x^2}}\,\mathrm{d}x=-\mathrm{d}u$,故 $\displaystyle\int\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x=\int-\mathrm{d}u=-u+c=-\sqrt{1-x^2}$,$\displaystyle\int\frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x=\int\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x+\int\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x=-\sqrt{1-x^2}+\sin^{-1}x+c$。

10. 令 $u=\tan^2 x$，則 $\mathrm{d}u=\sec^2 x\,\mathrm{d}x$，故 $\displaystyle\int\frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,\mathrm{d}x=\int\frac{1}{\sqrt{1-\tan^2 x}}\cdot\sec^2 x\,\mathrm{d}x=\int\frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u=\sin^{-1}u+c=\sin^{-1}(\tan x)+c$

11. $\displaystyle\int\sin^4 x\cos^5 x\,\mathrm{d}x=\int\sin^4 x\cdot\cos^4 x\cdot\cos x\,\mathrm{d}x=\int\sin^4 x\cdot(1-\sin^2 x)^2\cdot\cos x\,\mathrm{d}x=\int\Big(\sin^4 x-2\sin^6 x+\sin^8 x\Big)\cdot\cos x\,\mathrm{d}x$。令 $u=\sin x$，則 $\mathrm{d}u=\cos x\,\mathrm{d}x$，故 $\displaystyle\int\Big(\sin^4 x-2\sin^6 x+\sin^8 x\Big)\cdot\cos x\,\mathrm{d}x=\int\Big(u^4-2u^6+u^8\Big)\,\mathrm{d}u=\frac{u^5}{5}-\frac{2u^7}{7}+\frac{u^9}{9}+c=\frac{\sin^5 x}{5}-\frac{2\sin^7 x}{7}+\frac{\sin^9 x}{9}+c$

12. $\displaystyle\int\sin^2 x\cos^2 x\,\mathrm{d}x=\int\frac{1-\cos 2x}{2}\cdot\frac{1+\cos 2x}{2}\,\mathrm{d}x=\frac{1}{4}\int\Big(1-\cos^2 2x\Big)\,\mathrm{d}x=\frac{1}{4}\int\Big(1-\frac{1+\cos 4x}{2}\Big)\,\mathrm{d}x=\frac{1}{4}\int\Big(\frac{1}{2}+\frac{\cos 4x}{2}\Big)\,\mathrm{d}x=\frac{x}{8}-\frac{\sin 4x}{32}+c$

13. $\displaystyle\int\sin^{-\frac{2}{3}}x\cos^3 x\,\mathrm{d}x=\int\sin^{-\frac{2}{3}}x\cdot\cos^2 x\cdot\cos x\,\mathrm{d}x=\int\sin^{-\frac{2}{3}}x\cdot(1-\sin^2 x)\cdot\cos x\,\mathrm{d}x=\int\Big(\sin^{-\frac{2}{3}}x-\sin^{\frac{4}{3}}x\Big)\cdot\cos x\,\mathrm{d}x$。令 $u=\sin x$，則 $\mathrm{d}u=\cos x\,\mathrm{d}x$，故 $\displaystyle\int\Big(\sin^{-\frac{2}{3}}x-\sin^{\frac{4}{3}}x\Big)\cdot\cos x\,\mathrm{d}x=\int\Big(u^{-\frac{2}{3}}-u^{\frac{4}{3}}\Big)\,\mathrm{d}u=3u^{\frac{1}{3}}-\frac{3}{7}u^{\frac{7}{3}}+c=3\sin^{\frac{1}{3}}x-\frac{3}{7}\sin^{\frac{7}{3}}x+c$

14. 令 $u=\ln x$,則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$,故 $\displaystyle\int\frac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x=\int\sin^3(\ln x)\cos^3(\ln x)\cdot\frac{1}{x}\,\mathrm{d}x=\int\sin^3 u\cos^3 u\,\mathrm{d}u=\int\sin^3 u\cdot\cos^2 u\cdot\cos u\,\mathrm{d}u=\int\sin^3 u\Big(1-\sin^2 u\Big)\cdot\cos u\,\mathrm{d}u=\int\Big(\sin^3 u-\sin^5 u\Big)\cdot\cos u\,\mathrm{d}u$。令 $w=\sin u$，則 $\mathrm{d}w=\cos u\,\mathrm{d}u$,故 $\displaystyle\int\Big(\sin^3 u-\sin^5 u\Big)\cdot\cos u\,\mathrm{d}u=\int\Big(w^3-w^5\Big)\,\mathrm{d}w=\frac{w^4}{4}-\frac{w^6}{6}+c=\frac{\sin^4(\ln x)}{4}-\frac{\sin^6(\ln x)}{6}+c$

15. $\displaystyle\int\frac{\sin^3 x}{\cos^4 x}\,\mathrm{d}x=\int\frac{\sin^2 x\cdot\sin x}{\cos^4 x}\,\mathrm{d}x=\int\frac{\sin^2 x}{\cos^4 x}\cdot\sin x\,\mathrm{d}x=\int\frac{1-\cos^2 x}{\cos^4 x}\cdot\sin x\,\mathrm{d}x=\int\Big(\frac{1}{\cos^4 x}-\frac{1}{\cos^2 x}\Big)\cdot\sin x\,\mathrm{d}x$。令 $u=\cos x$，則 $\mathrm{d}u=-\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x=-\mathrm{d}u$，故 $\displaystyle\int\Big(\frac{1}{\cos^4 x}-\frac{1}{\cos^2 x}\Big)\cdot\sin x\,\mathrm{d}x=\int\Big(\frac{1}{u^4}-\frac{1}{u^2}\Big)(-)\,\mathrm{d}u=\int\Big(\frac{1}{u^2}-\frac{1}{u^4}\Big)\,\mathrm{d}u=-\frac{1}{u}+\frac{1}{3u^3}+c=-\frac{1}{\cos x}+\frac{1}{3\cos^3 x}+c$

16. 由積化和差公式 $\cos\alpha\cos\beta=\dfrac{1}{2}\Big(\cos(\alpha-\beta)+\cos(\alpha+\beta)\Big)$，$\displaystyle\int\cos ax\cos bx\,\mathrm{d}x=\int\frac{1}{2}\Big(\cos(ax-bx)+\cos(ax+bx)\Big)\,\mathrm{d}x=\int\frac{1}{2}\Big(\cos(a-b)x+\cos(a+b)x\Big)\,\mathrm{d}x=\frac{\sin(a-b)x}{2(a-b)}+\frac{\sin(a+b)x}{2(a+b)}+c$。

17. 由積化和差公式 $\sin\alpha\sin\beta=\dfrac{1}{2}\Big(\cos(\alpha-\beta)-\cos(\alpha+\beta)\Big)$，$\displaystyle\int\sin ax\sin bx\,\mathrm{d}x=\int\frac{1}{2}\Big(\cos(ax-bx)-\cos(ax+bx)\Big)\,\mathrm{d}x=\int\frac{1}{2}\Big(\cos(a-b)x-\cos(a+b)x\Big)\,\mathrm{d}x=\frac{\sin(a-b)x}{2(a-b)}-\frac{\sin(a+b)x}{2(a+b)}+c$。

18. 由積化和差公式 $\sin\alpha\cos\beta = \frac{1}{2}\Big(\sin(\alpha-\beta)+\sin(\alpha+\beta)\Big)$，$\int \sin ax\cos bx\,\mathrm{d}x = \int \frac{1}{2}\Big(\sin(ax-bx)+\sin(ax+bx)\Big)\,\mathrm{d}x = \int \frac{1}{2}\Big(\sin(a-b)x+\sin(a+b)x\Big)\,\mathrm{d}x = -\frac{\cos(a-b)x}{2(a-b)} - \frac{\cos(a+b)x}{2(a+b)} + c$。

部份積分法

結論. $\frac{\mathrm{d}}{\mathrm{d}x}\left(u(x)\,v(x)\right) = u(x)\,\frac{\mathrm{d}v(x)}{\mathrm{d}x} + v(x)\,\frac{\mathrm{d}u(x)}{\mathrm{d}x} \implies u(x)\,v(x) = \int u(x)\,\mathrm{d}v(x) + \int v(x)\,\mathrm{d}u(x)$
$\implies \int u\,\mathrm{d}v = u\,v - \int v\,\mathrm{d}u$。

例. 求 $\int x\,e^x\,\mathrm{d}x$。

解. 令 $u = x$，則 $\mathrm{d}u = \mathrm{d}x$。令 $\mathrm{d}v = e^x\,\mathrm{d}x$，則 $v = e^x$。故 $\int x\,e^x\,\mathrm{d}x = x\cdot e^x - \int e^x\,\mathrm{d}x = xe^x - e^x + c$。

例. 求 $\int \ln x\,\mathrm{d}x$。

解. 令 $u = \ln x$，則 $\mathrm{d}u = \frac{1}{x}\mathrm{d}x$。令 $\mathrm{d}v = \mathrm{d}x$，則 $v = x$。故 $\int \ln x\,\mathrm{d}x = \ln x\cdot x - \int x\cdot\frac{1}{x}\,\mathrm{d}x = x\ln x - x + c$。

例. 求 $\int x\tan^{-1}x\,\mathrm{d}x$。

解. 令 $u = \tan^{-1}x$，則 $\mathrm{d}u = \frac{1}{1+x^2}\mathrm{d}x$。令 $\mathrm{d}v = x\,\mathrm{d}x$，則 $v = \frac{x^2}{2}$。故 $\int x\tan^{-1}x\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^2}{2} - \frac{1}{2}\int\frac{x^2}{1+x^2}\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^2}{2} - \frac{1}{2}\int\Big(1-\frac{1}{1+x^2}\Big)\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^2}{2} - \frac{1}{2}\Big(x-\tan^{-1}x\Big) + c$。

例. 求 $\int x^2\sin x\,\mathrm{d}x$。

解. 令 $u = x^2$,則 $\mathrm{d}u = 2x\,\mathrm{d}x$。令 $\mathrm{d}v = \sin x\,\mathrm{d}x$,則 $v = -\cos x$。故 $\int x^2\,\sin x\,\mathrm{d}x = x^2\cdot(-\cos x)+2\int x\cos x\,\mathrm{d}x$。求 $\int x\cos x\,\mathrm{d}x$, 令 $u = x$, 則 $\mathrm{d}u = \mathrm{d}x$。令 $\mathrm{d}v = \cos x\,\mathrm{d}x$, 則 $v = \sin x$。故 $\int x\cos x\,\mathrm{d}x = x\cdot\sin x - \int\sin x\,\mathrm{d}x = x\cdot\sin x + \cos x$。由上，$\int x^2\sin x\,\mathrm{d}x = -x^2\cos x + 2(x\sin x + \cos x) + c$。

例. 求 $\int e^{ax}\cos bx\,\mathrm{d}x$ 與 $\int e^{ax}\sin bx\,\mathrm{d}x$，$a$、$b\in\mathbb{R}\setminus\{0\}$。

解. 求 $\int e^{ax}\cos bx\,\mathrm{d}x$, 令 $u = \cos bx$, 則 $\mathrm{d}u = -b\sin bx\,\mathrm{d}x$。令 $\mathrm{d}v = e^{ax}\mathrm{d}x$, 則 $v = \frac{1}{a}e^{ax}$。故 $\int e^{ax}\cos bx\,\mathrm{d}x = \cos bx\cdot\frac{1}{a}e^{ax} + \frac{b}{a}\int e^{ax}\cdot\sin bx\,\mathrm{d}x$。$(\star)$ 求 $\int e^{ax}\sin bx\,\mathrm{d}x$，令 $u = \sin bx$，則 $\mathrm{d}u = b\cos bx\,\mathrm{d}x$。令 $\mathrm{d}v = e^{ax}\mathrm{d}x$，則 $v = \frac{1}{a}e^{ax}$。故 $\int e^{ax}\sin bx\,\mathrm{d}x = \sin bx\cdot\frac{1}{a}e^{ax} - \frac{b}{a}\int e^{ax}\cdot\cos bx\,\mathrm{d}x$。$(\star\star)$ 令 $X = \int e^{ax}\cos bx\,\mathrm{d}x$, $Y = \int e^{ax}\sin bx\,\mathrm{d}x$，則由 $(\star)$，$(\star\star)$ $X = \cos bx\cdot\frac{1}{a}e^{ax} + \frac{b}{a}Y$，$Y = \sin bx\cdot\frac{1}{a}e^{ax} - \frac{b}{a}X$。解 X、Y 得 $X = \int e^{ax}\cos bx\,\mathrm{d}x = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$，$Y = \int e^{ax}\sin bx\,\mathrm{d}x = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$。

列表法

結論.

1. 將要微分函數寫左邊、積分函數寫右邊；左邊連續微分、右邊連續積分
2. 依序左上連右下函數相乘，符號正負相間；最下列寫成兩邊函數相乘的積分
3. 將上式所得項全部加總即為所求積分

例. 求 $\int (x+3)\, e^{2x}\, \mathrm{d}x$。

解.

Diff		Int
$x+3$	$+$	e^{2x}
1	$-$	$\frac{1}{2}e^{2x}$
0		$\frac{1}{4}e^{2x}$

$$\int (x+3)\, e^{2x}\, \mathrm{d}x = (x+3)\,\frac{1}{2}e^{2x} - \frac{1}{4}e^{2x}$$

例. 求 $\int (x^2-2x)\, e^{kx}\, \mathrm{d}x$。

解.

Diff		Int
x^2-2x	$+$	e^{kx}
$2(x-1)$	$-$	$\frac{1}{k}e^{kx}$
2	$+$	$\frac{1}{k^2}e^{kx}$
0		$\frac{1}{k^3}e^{kx}$

$$\int (x^2-2x)\, e^{kx}\, \mathrm{d}x = (x^2-2x)\,\frac{1}{k}e^{kx} - 2(x-1)\frac{1}{k^2}e^{kx} + 2\frac{1}{k^3}e^{kx}$$

例. 求 $\int x^4\, \sin 2x\, \mathrm{d}x$。

解.

Diff		Int
x^4	$+$	$\sin 2x$
$4x^3$	$-$	$-\frac{1}{2}\cos 2x$
$12x^2$	$+$	$-\frac{1}{4}\sin 2x$
$24x$	$-$	$\frac{1}{8}\cos 2x$
24	$+$	$\frac{1}{16}\sin 2x$
0		$-\frac{1}{32}\cos 2x$

$$\begin{aligned}
&\int x^4\, \sin 2x\, \mathrm{d}x \\
&= -x^4\,\frac{1}{2}\cos 2x + 4x^3\,\frac{1}{4}\sin 2x + 12x^2\,\frac{1}{8}\cos 2x - 24x\,\frac{1}{16}\sin 2x - 24\,\frac{1}{32}\cos 2x \\
&= \left(-\frac{x^4}{2} + \frac{3x^2}{2} - \frac{3}{4}\right)\cos 2x + \left(x^3 - \frac{3x}{2}\right)\sin 2x
\end{aligned}$$

例. 求 $\int x^5\, e^{ax}\, \mathrm{d}x$。

解.

Diff		Int
x^5	$+$	e^{ax}
$5x^4$	$-$	$\frac{1}{a}e^{ax}$
$20x^3$	$+$	$\frac{1}{a^2}e^{ax}$
$60x^2$	$-$	$\frac{1}{a^3}e^{ax}$
$120x$	$+$	$\frac{1}{a^4}e^{ax}$
120	$-$	$\frac{1}{a^5}e^{ax}$
0		$\frac{1}{a^6}e^{ax}$

$$\begin{aligned}
&\int x^5\, e^{ax}\, \mathrm{d}x \\
&= x^5\,\frac{1}{a}e^{ax} - 5x^4\,\frac{1}{a^2}e^{ax} + 20x^3\,\frac{1}{a^3}e^{ax} - 60x^2\,\frac{1}{a^4}e^{ax} + 120x\,\frac{1}{a^5}e^{ax} - 120\,\frac{1}{a^6}e^{ax} \\
&= \left(\frac{x^5}{a} - \frac{5x^4}{a^2} + \frac{20x^3}{a^3} - \frac{60x^2}{a^4} + \frac{120x}{a^5} - \frac{120}{a^6}\right) e^{ax}
\end{aligned}$$

例. 求 $\int (\sin^{-1} x)^2 \,\mathrm{d}x$。

解.

Diff		Int
$(\sin^{-1} x)^2$	$+$	1
$\frac{2\sin^{-1} x}{\sqrt{1-x^2}}$	$-$	x

Diff		Int
$2\sin^{-1} x$	$+$	$\frac{x}{\sqrt{1-x^2}}$
$\frac{2}{\sqrt{1-x^2}}$	$-$	$-\sqrt{1-x^2}$

$$\begin{aligned}
\int (\sin^{-1} x)^2 \,\mathrm{d}x &= (\sin^{-1} x)^2 \cdot x - \int \frac{2\sin^{-1} x}{\sqrt{1-x^2}} \cdot x \,\mathrm{d}x \\
&= (\sin^{-1} x)^2 \cdot x - \int 2\sin^{-1} x \cdot \frac{x}{\sqrt{1-x^2}} \,\mathrm{d}x \\
&= (\sin^{-1} x)^2 \cdot x - \left(-2\sin^{-1} x \cdot \sqrt{1-x^2} + \int \frac{2}{\sqrt{1-x^2}} \cdot \sqrt{1-x^2} \,\mathrm{d}x \right) \\
&= (\sin^{-1} x)^2 \cdot x + 2\sin^{-1} x \cdot \sqrt{1-x^2} - 2x
\end{aligned}$$

例. 求 $\int e^{ax} \cos bx \,\mathrm{d}x$。

解.

Diff		Int
$\cos bx$	$+$	e^{ax}
$-b\sin bx$	$-$	$\frac{1}{a}e^{ax}$
$-b^2\cos bx$	$+$	$\frac{1}{a^2}e^{ax}$

$$\begin{aligned}
&\int e^{ax} \cos bx \,\mathrm{d}x = \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} e^{ax} \sin bx - \frac{b^2}{a^2} \int e^{ax} \cos bx \,\mathrm{d}x \\
&\Longrightarrow \int e^{ax} \cos bx \,\mathrm{d}x = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}
\end{aligned}$$

例. 求 $\int e^{ax} \sin bx \,\mathrm{d}x$。

解.

Diff		Int
$\sin bx$	$+$	e^{ax}
$b\cos bx$	$-$	$\frac{1}{a}e^{ax}$
$-b^2\sin bx$	$+$	$\frac{1}{a^2}e^{ax}$

$$\begin{aligned}
&\int e^{ax} \sin bx \,\mathrm{d}x = \frac{1}{a} e^{ax} \sin bx - \frac{b}{a^2} e^{ax} \cos bx - \frac{b^2}{a^2} \int e^{ax} \sin bx \,\mathrm{d}x \\
&\Longrightarrow \int e^{ax} \sin bx \,\mathrm{d}x = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}
\end{aligned}$$

習題. 以部份積分法求下列不定積分。注意：可能會因為常數項而跟此處答案不同。

1. $\int x(\ln x)^3 \,\mathrm{d}x = \frac{x^2}{2}\left((\ln x)^3 - \frac{3(\ln x)^2}{2} + \frac{3\ln x}{2} - \frac{3}{4}\right) + c$
2. $\int x^2 \tan^{-1} x \,\mathrm{d}x = \frac{x^3}{3}\tan^{-1} x - \frac{x^2}{6} + \frac{\ln(1+x^2)}{6} + c$
3. $\int x^5 e^{-x^2} \,\mathrm{d}x = -\frac{1}{2} e^{-x^2}(x^4 + 2x^2 + 2) + c$
4. $\int x e^{\sqrt{x}} \,\mathrm{d}x = 2\,e^{\sqrt{x}}\left(x\sqrt{x} - 3x + 6\sqrt{x} - 6\right) + c$
5. $\int \sin(\ln x) \,\mathrm{d}x = \frac{x\,(\sin(\ln x) - \cos(\ln x))}{2} + c$
6. $\int \frac{\ln x}{\sqrt{1+x}} \,\mathrm{d}x = \ln x \cdot 2\sqrt{1+x} - 4\sqrt{1+x} - 2\ln(\sqrt{1+x} - 1) + 2\ln(\sqrt{1+x} + 1) + c$

解.

1. 令 $u = (\ln x)^3$，則 $\mathrm{d}u = 3(\ln x)^2 \cdot \frac{1}{x} \,\mathrm{d}x$；$\mathrm{d}v = x \,\mathrm{d}x$，則 $v = \frac{x^2}{2}$。故 $\int x(\ln x)^3 \,\mathrm{d}x = (\ln x)^3 \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot 3(\ln x)^2 \cdot \frac{1}{x} \,\mathrm{d}x = (\ln x)^3 \cdot \frac{x^2}{2} - \frac{3}{2}\int x(\ln x)^2 \,\mathrm{d}x$。令 $u = (\ln x)^2$，則 $\mathrm{d}u = 2\,\ln x \cdot \frac{1}{x} \,\mathrm{d}x$；$\mathrm{d}v = x \,\mathrm{d}x$，則 $v = \frac{x^2}{2}$。故 $\int x(\ln x)^2 \,\mathrm{d}x = (\ln x)^2 \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot 2\ln x \cdot \frac{1}{x} \,\mathrm{d}x = (\ln x)^2 \cdot \frac{x^2}{2} - \int x \ln x \,\mathrm{d}x$。令 $u = \ln x$，則 $\mathrm{d}u = \frac{1}{x} \,\mathrm{d}x$；

$\mathrm{d}v = x\,\mathrm{d}x$, 則 $v = \dfrac{x^2}{2}$。故 $\displaystyle\int x\ln x\,\mathrm{d}x = \ln x\cdot\frac{x^2}{2} - \int \frac{x^2}{2}\cdot\frac{1}{x}\,\mathrm{d}x = \ln x\cdot\frac{x^2}{2} - \frac{1}{2}\int x\,\mathrm{d}x = \ln x\cdot\frac{x^2}{2} - \frac{x^2}{4}$。以上，$\displaystyle\int x(\ln x)^3\,\mathrm{d}x = (\ln x)^3\cdot\frac{x^2}{2} - \frac{3}{2}\Big((\ln x)^2\cdot\frac{x^2}{2} - \Big(\ln x\cdot\frac{x^2}{2} - \frac{x^2}{4}\Big)\Big) = \frac{x^2}{2}\Big((\ln x)^3 - \frac{3(\ln x)^2}{2} + \frac{3\ln x}{2} - \frac{3}{4}\Big) + c$

2. 令 $u = \tan^{-1}x$，則 $\mathrm{d}u = \dfrac{1}{1+x^2}\,\mathrm{d}x$；$\mathrm{d}v = x^2\,\mathrm{d}x$，則 $v = \dfrac{x^3}{3}$。故 $\displaystyle\int x^2\tan^{-1}x\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^3}{3} - \int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x$。又 $\displaystyle\int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x = \frac{1}{3}\int\frac{(x^3+x)-x}{x^2+1}\,\mathrm{d}x = \frac{1}{3}\int\frac{x(x^2+1)-x}{x^2+1}\,\mathrm{d}x = \frac{1}{3}\int\Big(x - \frac{x}{x^2+1}\Big)\,\mathrm{d}x$。令 $w = x^2+1$，則 $\mathrm{d}w = 2x\,\mathrm{d}x \implies x\,\mathrm{d}x = \dfrac{1}{2}\,\mathrm{d}w$，故 $\displaystyle\int\frac{x}{x^2+1}\,\mathrm{d}x = \int\frac{1}{x^2+1}\cdot x\,\mathrm{d}x = \int\frac{1}{w}\cdot\frac{1}{2}\,\mathrm{d}w = \frac{1}{2}\ln w + c = \frac{1}{2}\ln(x^2+1) + c$。以上，$\displaystyle\int x^2\tan^{-1}x\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^3}{3} - \int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x = \tan^{-1}x\cdot\frac{x^3}{3} - \frac{x^2}{6} + \frac{\ln(x^2+1)}{6} + c$。

3. 令 $w = x^2$, 則 $\mathrm{d}w = 2x\,\mathrm{d}x \implies x\,\mathrm{d}x = \dfrac{1}{2}\,\mathrm{d}w$, 故 $\displaystyle\int x^5e^{-x^2}\,\mathrm{d}x = \int e^{-x^2}\cdot(x^2)^2\cdot x\,\mathrm{d}x = \int e^{-w}\cdot w^2\cdot\frac{1}{2}\,\mathrm{d}w = \frac{1}{2}\int w^2\,e^{-w}\,\mathrm{d}w = -\frac{1}{2}\,e^{-w}(w^2+2w+2) = -\frac{1}{2}\,e^{-x^2}(x^4+2x^2+2) + c$。

Diff		Int
w^2	$+$	e^{-w}
$2w$	$-$	$-e^{-w}$
2	$+$	e^{-w}
0		$-e^{-w}$

$$\int w^2e^{-w}\,\mathrm{d}w = -w^2\,e^{-w} - 2w\,e^{-w} - 2\,e^{-w} = -e^{-w}(w^2+2w+2)$$

4. 令 $w = \sqrt{x}$，則 $w^2 = x$，$\mathrm{d}x = 2w\,\mathrm{d}w$，故 $\displaystyle\int xe^{\sqrt{x}}\,\mathrm{d}x = \int w^2e^w\cdot 2w\,\mathrm{d}w = 2\int w^3e^w\,\mathrm{d}w = 2\,e^w(w^3 - 3w^2 + 6w - 6) = 2\,e^{\sqrt{x}}\,(x\sqrt{x} - 3x + 6\sqrt{x} - 6) + c$

Diff		Int
w^3	$+$	e^w
$3w^2$	$-$	e^w
$6w$	$+$	e^w
6	$-$	e^w
0		e^w

$$\int w^3\,e^w\,\mathrm{d}w = w^3\,e^w - 3w^2\,e^w + 6w\,e^w - 6\,e^w = e^w(w^3 - 3w^2 + 6w - 6)$$

5. 令 $w = \ln x$，則 $e^w = x$，$\mathrm{d}x = e^w\,\mathrm{d}w$，故 $\displaystyle\int\sin(\ln x)\,\mathrm{d}x = \int\sin w\cdot e^w\,\mathrm{d}w = \int e^w\,\sin w\,\mathrm{d}w = \frac{e^w(\sin w - \cos w)}{2} = \frac{x\,(\sin(\ln x) - \cos(\ln x))}{2} + c$

Diff		Int
$\sin w$	$+$	e^w
$\cos w$	$-$	e^w
$-\sin w$	$+$	e^w

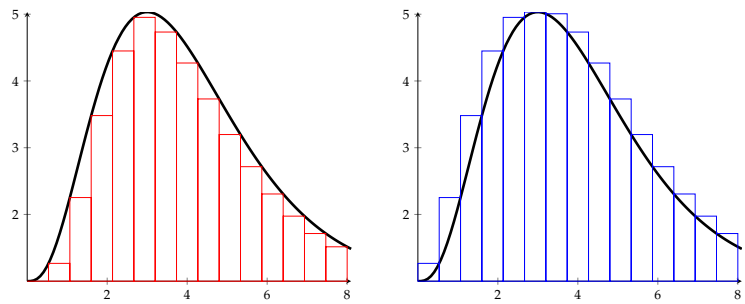
$$\int e^w\sin w\,\mathrm{d}w = e^w\sin w - e^w\cos w - \int e^w\sin w\,\mathrm{d}w$$

$$\implies \int e^w\sin w\,\mathrm{d}w = \frac{e^w(\sin w - \cos w)}{2}$$

6. 令 $u = \ln x$，則 $\mathrm{d}u = \dfrac{1}{x}\,\mathrm{d}x$；$\mathrm{d}v = \dfrac{1}{\sqrt{1+x}}\,\mathrm{d}x$，則 $v = 2\sqrt{1+x}$。故 $\displaystyle\int\frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x = \ln x\cdot 2\sqrt{1+x} - 2\int\frac{\sqrt{1+x}}{x}\,\mathrm{d}x$。令 $w = \sqrt{1+x}$，則 $w^2 = 1+x \implies x = w^2-1$，$2w\,\mathrm{d}w = \mathrm{d}x$，故 $\displaystyle\int\frac{\sqrt{1+x}}{x}\,\mathrm{d}x = \int\frac{w}{w^2-1}\cdot 2w\,\mathrm{d}w = 2\int\frac{w^2}{w^2-1}\,\mathrm{d}w = 2\int\frac{(w^2-1)+1}{w^2-1}\,\mathrm{d}w = 2w + 2\int\frac{1}{w^2-1}\,\mathrm{d}w = 2\sqrt{1+x} +$

$2\int \frac{1}{w^2-1}\,\mathrm{d}w$。由部份分式 $\frac{1}{w^2-1}=\frac{1}{2}\left(\frac{1}{w-1}-\frac{1}{w+1}\right)$，$\int \frac{1}{w^2-1}\,\mathrm{d}w=\frac{1}{2}\int\left(\frac{1}{w-1}-\frac{1}{w+1}\right)\mathrm{d}w=\frac{1}{2}\Big(\ln(w-1)-\ln(w+1)\Big)=\frac{1}{2}\Big(\ln(\sqrt{1+x}-1)-\ln(\sqrt{1+x}+1)\Big)$。以上，$\int \frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x=\ln x\cdot 2\sqrt{1+x}-2\left(2\sqrt{1+x}+2\left(\frac{1}{2}\Big(\ln(\sqrt{1+x}-1)-\ln(\sqrt{1+x}+1)\Big)\right)\right)=\ln x\cdot 2\sqrt{1+x}-4\sqrt{1+x}-2\ln(\sqrt{1+x}-1)+2\ln(\sqrt{1+x}+1)+c$。

4.2 定積分



定義. 令 f 在 $[a,b]$ 連續。

- $\mathbb{P}: a=x_0<x_1<x_2<\cdots<x_n=b$ 為 $[a,b]$ 的分割。
- $\Delta x_k=x_k-x_{k-1}$，$\|\mathbb{P}\|=\max\{\,|\Delta x_k|\mid 1\leqslant k\leqslant n\}$，$x_{k-1}\leqslant \xi_k\leqslant x_k$，$k=1,2,\ldots,n$
- $u_k=\sup\{\,f(x)\mid x_{k-1}\leqslant x\leqslant x_k\}$，$l_k=\inf\{\,f(x)\mid x_{k-1}\leqslant x\leqslant x_k\}$，$k=1,2,\ldots,n$
- $R(f,\mathbb{P})=\sum_{k=1}^{n} f(\xi_k)\,\Delta x_k$，$U(f,\mathbb{P})=\sum_{k=1}^{n} u_k\,\Delta x_k$，$L(f,\mathbb{P})=\sum_{k=1}^{n} l_k\,\Delta x_k$；
  顯然 $L(f,\mathbb{P})\leqslant R(f,\mathbb{P})\leqslant U(f,\mathbb{P})$。
- $\int_a^b f(x)\,\mathrm{d}x\equiv \lim\limits_{\substack{n\to\infty\\ \|\mathbb{P}\|\to 0}} R(f,\mathbb{P})$；若此極限存在，稱 f 在 $[a,b]$ 可積（分）。

性質 (定積分基本性質)**.** 令 f、g 在包含 a、b、c 之區間為可積，α、$\beta\in\mathbb{R}$。則

1. $\int_a^a f(x)\,\mathrm{d}x=0$
2. $\int_a^b f(x)\,\mathrm{d}x=-\int_b^a f(x)\,\mathrm{d}x$
3. $\int_a^b \Big(\alpha\, f(x)+\beta\, g(x)\Big)\,\mathrm{d}x=\alpha\int_a^b f(x)\,\mathrm{d}x+\beta\int_a^b g(x)\,\mathrm{d}x$
4. $\int_a^b f(x)\,\mathrm{d}x+\int_b^c f(x)\,\mathrm{d}x=\int_a^c f(x)\,\mathrm{d}x$
5. $\int_a^b f(x)\,\mathrm{d}x\leqslant\int_a^b g(x)\,\mathrm{d}x$，若 $f(x)\leqslant g(x)\ \forall\, x\in[a,b]$，$a\leqslant b$
6. $\left|\int_a^b f(x)\,\mathrm{d}x\right|\leqslant\int_a^b |\,f(x)\,|\,\mathrm{d}x,\quad a\leqslant b$
7. $\int_{-a}^a f(x)\,\mathrm{d}x=0$，若 $f(x)$ 為奇函數

8. $\int_{-a}^{a} f(x)\,\mathrm{d}x = 2\int_{0}^{a} f(x)\,\mathrm{d}x$，若 $f(x)$ 為偶函數

例.

1. $\int_{-\sqrt{2}}^{\sqrt{2}} \sqrt{2-x^2}\,\mathrm{d}x = \pi$（半徑 $\sqrt{2}$ 之半圓面積）
2. $\int_{-1}^{2} \operatorname{sgn}(x)\,\mathrm{d}x = 2\cdot 1 - 1\cdot 1 = 1$
3. $\int_{-\pi}^{\pi} \sin(x^7-5x^3)\,\mathrm{d}x = 0$（$\sin(x^7-5x^3)$ 為奇函數）
4. $\int_{-6}^{6} e^{-x^4}\sin(\sin x)\,\mathrm{d}x = 0$（$e^{-x^4}\sin(\sin x)$ 為奇函數）
5. $\int_{-4}^{4} \left(e^{x}-e^{-x}\right)\mathrm{d}x = 0$（$e^{x}-e^{-x}$ 為奇函數）
6. $\int_{-100}^{100} \left(e^{9x^5-2x^7}-e^{-9x^5+2x^7}\right)\mathrm{d}x = 0$（$e^{9x^5-2x^7}-e^{-9x^5+2x^7}$ 為奇函數）
7. $\int_{-a}^{a} |x|\,\mathrm{d}x = 2\int_{0}^{a} |x|\,\mathrm{d}x = a^2$（$|x|$ 為奇函數；兩 $a\times a$ 直角三角形面積）
8. 定義 $\int_{1}^{x} \frac{1}{\tau}\,\mathrm{d}\tau \equiv \ln x$，則 $\int_{\frac{1}{4}}^{3} \frac{1}{x}\,\mathrm{d}x = \int_{\frac{1}{4}}^{1} \frac{1}{x}\,\mathrm{d}x + \int_{1}^{3} \frac{1}{x}\,\mathrm{d}x = -\int_{1}^{\frac{1}{4}} \frac{1}{x}\,\mathrm{d}x + \int_{1}^{3} \frac{1}{x}\,\mathrm{d}x = \ln 12$
9. $\int_{0}^{2} \sqrt{4-x^2}\cdot \operatorname{sgn}(x-1)\,\mathrm{d}x = \pi - \sqrt{3}$

結論. $\sum_{k=1}^{m} k^2 = \frac{m(m+1)(2m+1)}{6}$。

例. 求 $\int_{0}^{1} x^2\,\mathrm{d}x$。

解. 建立 $[0,\,1]$ 分割 $\mathbb{P}:\left\{0,\frac{1}{n},\frac{2}{n},\ldots,\frac{n}{n}\right\}$，則 $\Delta x_k = \frac{1}{n}\ \forall\, k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$。$f(x)=x^2$ 且 f 在 $[0,\,1]$ 嚴格遞增，$U(f,\mathbb{P}) = \sum_{k=1}^{n} u_k\,\Delta x_k = \sum_{k=1}^{n} \frac{k^2}{n^2}\cdot\frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n} k^2$，$L(f,\mathbb{P}) = \sum_{k=1}^{n} l_k\,\Delta x_k = \sum_{k=1}^{n} \frac{(k-1)^2}{n^2}\cdot\frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n-1} k^2$，$L(f,\mathbb{P}) \leqslant R(f,\mathbb{P}) \leqslant U(f,\mathbb{P})$。由 $\sum_{k=1}^{m} k^2 = \frac{m(m+1)(2m+1)}{6}$，$\lim_{n\to\infty} U(f,\mathbb{P}) = \lim_{n\to\infty} \frac{1}{n^3}\sum_{k=1}^{n} k^2 = \lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{3}$，$\lim_{n\to\infty} L(f,\mathbb{P}) = \lim_{n\to\infty} \frac{1}{n^3}\sum_{k=1}^{n-1} k^2 = \lim_{n\to\infty} \frac{(n-1)n(2n-1)}{6n^3} = \frac{1}{3}$，故由夾擊定理 $\lim_{n\to\infty} R(f,\mathbb{P}) = \frac{1}{3}$。

結論. 若 $r\in\mathbb{R}$，$\sum_{k=0}^{n} r^k = \frac{r^{n+1}-1}{r-1}$。

例. 求 $\int_{0}^{1} e^{x}\,\mathrm{d}x$。

解. 建立 $[0,\,1]$ 分割 $\mathbb{P}:\left\{0,\frac{1}{n},\frac{2}{n},\ldots,\frac{n}{n}\right\}$，則 $\Delta x_k = \frac{1}{n}\ \forall\, k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$。$f(x)=e^x$ 且 f 在 $[0,\,1]$ 嚴格遞增，$U(f,\mathbb{P}) = \sum_{k=1}^{n} u_k\,\Delta x_k = \sum_{k=1}^{n} e^{\frac{k}{n}}\cdot\frac{1}{n} = \frac{1}{n}\frac{e^{\frac{1}{n}}\left(e^{\frac{n}{n}}-1\right)}{e^{\frac{1}{n}}-1} = \frac{e-1}{n}\frac{e^{\frac{1}{n}}}{e^{\frac{1}{n}}-1}$，$L(f,\mathbb{P}) = \sum_{k=1}^{n} l_k\,\Delta x_k = \sum_{k=1}^{n} e^{\frac{k-1}{n}}\cdot\frac{1}{n} = \frac{1}{n}\frac{e^{0}\left(e^{\frac{n}{n}}-1\right)}{e^{\frac{1}{n}}-1} = \frac{e-1}{n}\frac{1}{e^{\frac{1}{n}}-1}$，$L(f,\mathbb{P}) \leqslant R(f,\mathbb{P}) \leqslant U(f,\mathbb{P})$。$\lim_{n\to\infty} U(f,\mathbb{P}) = \lim_{n\to\infty} \frac{e-1}{n}\frac{e^{\frac{1}{n}}}{e^{\frac{1}{n}}-1} =$

$(e-1)\lim\limits_{x\to 0+}\dfrac{xe^x}{e^x-1}=(e-1)\lim\limits_{x\to 0+}\dfrac{e^x+xe^x}{e^x}=e-1$，$\lim\limits_{n\to\infty}L(f,\mathbb{P})=\lim\limits_{n\to\infty}\dfrac{e-1}{n}\dfrac{1}{e^{\frac{1}{n}}-1}=(e-1)\lim\limits_{x\to 0+}\dfrac{x}{e^x-1}=(e-1)\lim\limits_{x\to 0+}\dfrac{1}{e^x}=e-1$，故由夾擊定理 $\lim\limits_{n\to\infty}R(f,\mathbb{P})=e-1$。

定理 (微積分基本定理（Fundamental Theorem of Calculus，FTC）).

1. 若 f 在 $[a,b]$ 連續，令 $F(x)=\displaystyle\int_a^x f(\tau)\,\mathrm{d}\tau$ 且 $a\leqslant x\leqslant b$，則 $F'(x)=f(x)\ \forall\, x\in[a,b]$。
2. 若 $G'(x)=f(x)\ \forall\, x\in[a,b]$，則 $\displaystyle\int_a^b f(x)\,\mathrm{d}x=G(b)-G(a)\equiv G(x)\Big|_a^b$。

註. 由 FTC，$\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 可由 f 的反導函數（不定積分）得出，不需繁複極限計算！

證.

1. WLOG 令 $h>0$。則 $\left|\dfrac{F(x+h)-F(x)}{h}-f(x)\right|=\left|\dfrac{1}{h}\displaystyle\int_x^{x+h}(f(\tau)-f(x))\,\mathrm{d}\tau\right|\leqslant\dfrac{1}{h}\displaystyle\int_x^{x+h}\Big|f(\tau)-f(x)\Big|\,\mathrm{d}\tau\leqslant\max\limits_{x\leqslant\tau\leqslant x+h}\Big|f(\tau)-f(x)\Big|\to 0$，當 $h\to 0$ 且 f 為連續。
2. $(G(x)-F(x))'=f(x)-f(x)=0\ \forall\, x\in[a,b]$，故 $G(x)-F(x)$ 在 $[a,b]$ 為常數 $\implies G(a)-F(a)=G(b)-F(b)\implies G(b)-G(a)=F(b)-F(a)=\displaystyle\int_a^b f(x)\,\mathrm{d}x$。

例 (由 FTC 求定積分).

1. x^2 之反導函數為 $\dfrac{x^3}{3}$，故 $\displaystyle\int_0^1 x^2\,\mathrm{d}x=\dfrac{1^3}{3}-\dfrac{0^3}{3}=\dfrac{1}{3}$。
2. e^x 之反導函數為 e^x，故 $\displaystyle\int_0^1 e^x\,\mathrm{d}x=e^1-e^0=e-1$。

結論. $\displaystyle\sum_{k=1}^n\cos kx=\frac{1}{2}\left(\frac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}-1\right)$

證. 考慮 $\displaystyle\sum_{k=1}^n e^{ikx}=\frac{e^{ix}(1-e^{inx})}{1-e^{ix}}$,則 $\displaystyle\sum_{k=1}^n\cos kx=\Re\Big\{\sum_{k=1}^n e^{ikx}\Big\}=\Re\Big\{\frac{e^{ix}(1-e^{inx})}{1-e^{ix}}\Big\}=\Re\Big\{\frac{e^{ix}\,e^{\frac{inx}{2}}\left(e^{-\frac{inx}{2}}-e^{\frac{inx}{2}}\right)}{e^{\frac{ix}{2}}\left(e^{-\frac{ix}{2}}-e^{\frac{ix}{2}}\right)}\Big\}=\dfrac{\cos\frac{(n+1)x}{2}\,\sin\frac{nx}{2}}{\sin\frac{x}{2}}=\dfrac{\sin\left(\frac{nx}{2}+\frac{(n+1)x}{2}\right)+\sin\left(\frac{nx}{2}+\frac{(n+1)x}{2}\right)}{2\,\sin\frac{x}{2}}=\dfrac{\sin\left(n+\frac{1}{2}\right)x-\sin\frac{x}{2}}{2\,\sin\frac{x}{2}}=\dfrac{1}{2}\left(\dfrac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}-1\right)$。

例. 求 $\displaystyle\int_0^{\frac{\pi}{2}}\cos x\,\mathrm{d}x$。

解.

- (極限定義) 建立 $\left[0,\dfrac{\pi}{2}\right]$ 分割 $\mathbb{P}:\left\{0,\dfrac{\pi}{2n},\dfrac{2\pi}{2n},\ldots,\dfrac{n\pi}{2n}\right\}$，則 $\Delta x_k=\dfrac{\pi}{2n}\ \forall\, k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$, 則積分為 $\displaystyle\lim_{n\to\infty}R(f,\mathbb{P})=\sum_{k=1}^n f(\xi_k)\,\Delta x_k=\lim_{n\to\infty}\sum_{k=1}^n\cos\frac{k\pi}{2n}\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\sum_{k=1}^n\cos\left(k\left(\frac{\pi}{2n}\right)\right)\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\frac{1}{2}\left(\frac{\sin\left(\left(n+\frac{1}{2}\right)\cdot\frac{\pi}{2n}\right)}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}-1\right)\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\frac{\sin\left(\left(n+\frac{1}{2}\right)\frac{\pi}{2n}\right)\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}=\lim_{n\to\infty}\sin\left(\frac{1}{2}+\frac{1}{4n}\right)\pi\cdot\lim_{n\to\infty}\frac{\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}=\sin\frac{\pi}{2}\cdot 1=1.$
- (FTC) $\cos x$ 之反導函數為 $\sin x$，故 $\displaystyle\int_0^{\frac{\pi}{2}}\cos x\,\mathrm{d}x=\sin\frac{\pi}{2}-\sin 0=1$。

性質. 令 $F(x)=\int_{v(x)}^{u(x)} f(\tau)\,\mathrm{d}\tau$，則 $F'(x)=f(u(x))\cdot u'(x)-f(v(x))\cdot v'(x)$。

證. 令 $a\in\mathbb{R}$，$F(x)=\int_{v(x)}^{u(x)} f(\tau)\,\mathrm{d}\tau=\int_a^{u(x)} f(\tau)\,\mathrm{d}\tau-\int_a^{v(x)} f(\tau)\,\mathrm{d}\tau$。令 $G(x)\equiv\int_a^x f(\tau)\,\mathrm{d}\tau$，則 $G'(x)=f(x)$，$F(x)=\int_a^{u(x)} f(\tau)\,\mathrm{d}\tau-\int_a^{v(x)} f(\tau)\,\mathrm{d}\tau=G(u(x))-G(v(x))$；故 $F'(x)=(G(u(x))-G(v(x)))'=G'(u(x))\cdot u'(x)-G'(v(x))\cdot v'(x)=f(u(x))\cdot u'(x)-f(v(x))\cdot v'(x)$。

例.

1. $F(x)=\int_1^x \frac{1}{1+\tau^4}\,\mathrm{d}\tau \implies F'(x)=\frac{1}{1+x^4}$
2. $F(x)=\int_2^{\sqrt{x}} \sin\tau\,\mathrm{d}\tau \implies F'(x)=\sin\sqrt{x}\cdot\frac{1}{2\sqrt{x}}$
3. $F(x)=\int_x^{2x} \tau^3\,\mathrm{d}\tau \implies F'(x)=(2x)^3\cdot 2-x^3\cdot 1=15x^3$

定積分變數變換

結論.

- 先求得反導函數後再代入：$\int_a^b f'(g(x))\,g'(x)\,\mathrm{d}x=f(g(x))\Big|_{x=a}^{x=b}=f(g(b))-f(g(a))$
- 變數變換並改變積分範圍：$\int_a^b f'(g(x))\,g'(x)\,\mathrm{d}x=\int_a^b f'(g(x))\,\mathrm{d}g(x)=\int_{g(a)}^{g(b)} f'(u)\,\mathrm{d}u=f(u)\Big|_{u=g(a)}^{u=g(b)}=f(g(b))-f(g(a))$

例. 求 $\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x$。

解.

- 先求得反導函數後再代入：令 $u=1+x^4$，則 $\mathrm{d}u=4x^3\,\mathrm{d}x \implies x^3\,\mathrm{d}x=\frac{\mathrm{d}u}{4}$，故 $\int x^3(1+x^4)^3\,\mathrm{d}x=\int(1+x^4)^3\,x^3\,\mathrm{d}x=\int u^3\,\frac{\mathrm{d}u}{4}=\frac{u^4}{16}+c=\frac{(1+x^4)^4}{16}+c$。故 $\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x=\frac{(1+x^4)^4}{16}\Big|_{x=0}^{x=1}=\frac{(1+1^4)^4-(1+0^4)^4}{16}=\frac{15}{16}$。
- 變數變換並改變積分範圍：令 $u=1+x^4$，則 $\mathrm{d}u=4x^3\,\mathrm{d}x \implies x^3\,\mathrm{d}x=\frac{\mathrm{d}u}{4}$。積分範圍 x 由 0 至 1，則變數變換後 u 由 $1+0^4=1$ 至 $1+1^4=2$，故 $\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x=\int_0^1(1+x^4)^3\,x^3\,\mathrm{d}x=\int_1^2 u^3\,\frac{\mathrm{d}u}{4}=\frac{1}{4}\int_1^2 u^3\,\mathrm{d}u=\frac{1}{16}u^4\Big|_{u=1}^{u=2}=\frac{2^4-1^4}{16}=\frac{15}{16}$。

例. 求 $\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x$。

解.

- 先求得反導函數後再代入：令 $u=1+x^2$，則 $\mathrm{d}u=2x\,\mathrm{d}x \implies 4x\,\mathrm{d}x=2\,\mathrm{d}u$，故 $\int\frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x=\int\frac{2}{\sqrt{u}}\,\mathrm{d}u=4\sqrt{u}+c=4\sqrt{1+x^2}+c$。故 $\int_0^{\sqrt{3}}\frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x=4\sqrt{1+x^2}\Big|_{x=0}^{x=\sqrt{3}}=4\sqrt{1+3}-4\sqrt{1}=4$。

- 變數變換並改變積分範圍：令 $u = 1 + x^2$，則 $\mathrm{d}u = 2x\,\mathrm{d}x \implies 4x\,\mathrm{d}x = 2\,\mathrm{d}u$。積分範圍 x 由 0 至 $\sqrt{3}$，則變數變換後 u 由 $1 + 0^2 = 1$ 至 $1 + (\sqrt{3})^2 = 4$，故 $\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = \int_1^4 \frac{2}{\sqrt{u}}\,\mathrm{d}u = 4\sqrt{u}\Big|_{u=1}^{u=4} = 4\,(\sqrt{4} - \sqrt{1}) = 4$。

例. 求 $\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x$。

解.

- 先求得反導函數後再代入：令 $u = \cos x$，則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$，故 $\int 3\cos^2 x \sin x\,\mathrm{d}x = -3\int u^2\,\mathrm{d}u = -u^3 + c = -\cos^3 x + c$。故 $\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x = -\cos^3 x\Big|_{x=0}^{x=\pi} = -(\cos^3\pi - \cos^3 0) = -((-1)^3 - 1^3) = 2$。
- 變數變換並改變積分範圍：令 $u = \cos x$，則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$。積分範圍 x 由 0 至 π，則變數變換後 u 由 $\cos 0 = 1$ 至 $\cos\pi = -1$，故 $\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x = -\int_1^{-1} 3u^2\,\mathrm{d}u = -u^3\Big|_{u=1}^{u=-1} = -((-1)^3 - 1^3) = 2$。

4.3 瑕積分

定義 (瑕積分).

- 若 (i) f 在 I 有無限函數值或 (ii) I 為無限區間，$\int_I f(x)\,\mathrm{d}x$ 稱為瑕積分 (improper integral)。
- $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \lim\limits_{\substack{a\to-\infty \\ b\to\infty}} \int_a^b f(x)\,\mathrm{d}x$。

例. 求 $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,\mathrm{d}x$。

解. $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,\mathrm{d}x = 2\int_0^{\infty} \frac{1}{1+x^2}\,\mathrm{d}x = 2\lim\limits_{b\to\infty}\int_0^b \frac{1}{1+x^2}\,\mathrm{d}x = 2\lim\limits_{b\to\infty}\tan^{-1} b = 2\cdot\frac{\pi}{2} = \pi$。

例. 求 $\int_1^{\infty} \frac{1}{x^2}\,\mathrm{d}x$。

解. $\int_1^{\infty} \frac{1}{x^2}\,\mathrm{d}x = \lim\limits_{b\to\infty}\int_1^b \frac{1}{x^2}\,\mathrm{d}x = \lim\limits_{b\to\infty}\Big(-\frac{1}{b} + 1\Big) = 1$

例. 求 $\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x$。

解. $\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim\limits_{c\to 0+}\int_c^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim\limits_{c\to 0+} 2\sqrt{x}\Big|_c^1 = \lim\limits_{c\to 0+}(2 - 2\sqrt{c}) = 2$

結論. 給定 $0 < a < \infty$。

- $\int_a^{\infty} \frac{1}{x^p}\,\mathrm{d}x$ 當 $p > 1$ 收斂至 $\frac{a^{1-p}}{p-1}$，當 $p \leqslant 1$ 發散至 ∞。
- $\int_0^{a} \frac{1}{x^p}\,\mathrm{d}x$ 當 $p < 1$ 收斂至 $\frac{a^{1-p}}{1-p}$，當 $p \geqslant 1$ 發散至 ∞。

結論. 令 $-\infty \leqslant a < b \leqslant \infty$，函數 f、g 在 (a, b) 連續，$0 \leqslant f(x) \leqslant g(x)\ \forall x$。

- 若 $\int_a^b g(x)\,\mathrm{d}x$ 收斂，$\int_a^b f(x)\,\mathrm{d}x$ 收斂。

- 若 $\int_a^b f(x)\,\mathrm{d}x$ 發散，$\int_a^b g(x)\,\mathrm{d}x$ 發散。

例. 證明 $\int_0^\infty e^{-x^2}\,\mathrm{d}x$ 收斂。

解. 由 $e^{-x^2} \leqslant 1\ \forall 0 \leqslant x < 1$ 及 $e^{-x^2} \leqslant e^{-x}\ \forall x \geqslant 1$，$\int_0^\infty e^{-x^2}\,\mathrm{d}x = \int_0^1 e^{-x^2}\,\mathrm{d}x + \int_1^\infty e^{-x^2}\,\mathrm{d}x \leqslant \int_0^1 1\,\mathrm{d}x + \int_1^\infty e^{-x}\,\mathrm{d}x = 1 + \frac{1}{e}$，故 $\int_0^\infty e^{-x^2}\,\mathrm{d}x$ 收斂。

例. 證明 $\forall x > 0$，$\Gamma(x) \equiv \int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂。

解. 由 $\lim\limits_{t\to\infty} t^{x-1}e^{-\frac{t}{2}} = 0$, $\exists T > 0$ 使 $t^{x-1}e^{-\frac{t}{2}} \leqslant 1\ \forall t \geqslant T$。則 $\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = \int_0^T t^{x-1}e^{-t}\,\mathrm{d}t + \int_T^\infty t^{x-1}e^{-t}\,\mathrm{d}t \leqslant \int_0^T t^{x-1}\,\mathrm{d}t + \int_T^\infty 1\cdot e^{-\frac{t}{2}}\,\mathrm{d}t = \frac{T^x}{x} + 2e^{-\frac{T}{2}}$。故 $\forall x > 0$；$\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂。

例. 證明 $\forall x > 0$，$\Gamma(x+1) = x\,\Gamma(x)$。

解. $\Gamma(x+1) = \int_0^\infty t^x e^{-t}\,\mathrm{d}t = \lim\limits_{\substack{a\to 0+\\ b\to\infty}} \int_a^b t^x e^{-t}\,\mathrm{d}t$。令 $u = t^x$，則 $\mathrm{d}u = xt^{x-1}\,\mathrm{d}t$。令 $\mathrm{d}v = e^{-t}\,\mathrm{d}t$，則 $v = -e^{-t}$。故 $\lim\limits_{\substack{a\to 0+\\ b\to\infty}} \int_a^b t^x e^{-t}\,\mathrm{d}t = \lim\limits_{\substack{a\to 0+\\ b\to\infty}} \left(-t^x e^{-t}\Big|_a^b + \int_a^b e^{-t}\cdot xt^{x-1}\,\mathrm{d}t \right) = 0 + x\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = x\,\Gamma(x)$。

例. 由 $\int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$，求 $\Gamma\left(\frac{1}{2}\right)$ 與 $\Gamma\left(\frac{3}{2}\right)$。

解.

1. $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}}e^{-t}\,\mathrm{d}t = \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t$。令 $u = \sqrt{t}$，則 $t = u^2$，$\mathrm{d}u = \frac{1}{2\sqrt{t}}\,\mathrm{d}t \implies \frac{1}{\sqrt{t}}\,\mathrm{d}t = 2\,\mathrm{d}u$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\sqrt{0} = 0$ 至 $\sqrt{\infty} = \infty$，故 $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t = 2\int_0^\infty e^{-u^2}\,\mathrm{d}u = 2\cdot\frac{\sqrt{\pi}}{2} = \sqrt{\pi}$。
2. 由 $\Gamma(x+1) = x\,\Gamma(x)$，$\Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2}+1\right) = \frac{1}{2}\,\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi}$。

例. 由 $\int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$ 且 $\alpha > 0$，求 $\int_0^\infty t\,e^{-\alpha t^2}\,\mathrm{d}t$ 與 $\int_0^\infty t^2\,e^{-\alpha t^2}\,\mathrm{d}t$。

解.

1. 令 $u = \alpha t^2$，則 $\mathrm{d}u = 2\alpha t\,\mathrm{d}t \implies t\,\mathrm{d}t = \frac{\mathrm{d}u}{2\alpha}$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\alpha\cdot 0^2 = 0$ 至 $\alpha\cdot\infty^2 = \infty$，故 $\int_0^\infty t\,e^{-\alpha t^2}\,\mathrm{d}t = \int_0^\infty e^{-u}\,\frac{\mathrm{d}u}{2\alpha} = \frac{1}{2\alpha}\int_0^\infty e^{-u}\,\mathrm{d}u = \frac{1}{2\alpha}\left(-e^{-u}\Big|_0^\infty \right) = \frac{1}{2\alpha}$。
2. 令 $u = \alpha t^2$，則 $t = \frac{\sqrt{u}}{\sqrt{\alpha}}$，$\mathrm{d}u = 2\alpha t\,\mathrm{d}t \implies t\,\mathrm{d}t = \frac{\mathrm{d}u}{2\alpha}$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\alpha\cdot 0^2 = 0$ 至 $\alpha\cdot\infty^2 = \infty$，故 $\int_0^\infty t^2\,e^{-\alpha t^2}\,\mathrm{d}t = \int_0^\infty t\,e^{-\alpha t^2}\,t\,\mathrm{d}t = \int_0^\infty \frac{\sqrt{u}}{\sqrt{\alpha}}\,e^{-u}\,\frac{\mathrm{d}u}{2\alpha} = \frac{1}{2\alpha^{\frac{3}{2}}}\int_0^\infty u^{\frac{1}{2}}\,e^{-u}\,\mathrm{d}u = \frac{1}{2\alpha^{\frac{3}{2}}}\cdot\Gamma\left(\frac{3}{2}\right) = \frac{1}{2\alpha^{\frac{3}{2}}}\cdot\frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{4\alpha^{\frac{3}{2}}}$。